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EMERGENCE OF GLOBALLY ATTRACTING FIXED POINTS IN DEEP NEURAL NETWORKS WITH NONLINEAR ACTIVATIONS

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ABSTRACT

Understanding how neural networks transform input data across layers is fundamental to unraveling their learning and generalization capabilities. Although prior work has used insights from kernel methods to study neural networks, a global analysis of how the similarity between hidden representations evolves across layers remains underexplored. In this paper, we introduce a theoretical framework for the evolution of the kernel sequence, which measures the similarity between the hidden representation for two different inputs. Operating under the mean-field regime, we show that the kernel sequence evolves deterministically via a kernel map, which only depends on the activation function. By expanding activation using Hermite polynomials and using their algebraic properties, we derive an explicit form for kernel map and fully characterize its fixed points. Our analysis reveals that for nonlinear activations, the kernel sequence converges globally to a unique fixed point, which can correspond to orthogonal or similar representations depending on the activation and network architecture. We further extend our results to networks with residual connections and normalization layers, demonstrating similar convergence behaviors. This work provides new insights into the implicit biases of deep neural networks and how architectural choices influence the evolution of representations across layers.

1 Introduction

Deep neural networks have revolutionized various fields, from computer vision to natural language processing, due to their remarkable ability to learn complex patterns from data. Understanding the internal mechanisms that govern their learning and generalization capabilities remains a fundamental challenge.

One approach to studying these transformations is through the lens of kernel methods. Kernel methods have a long history in machine learning for analyzing relationships between data points in high-dimensional spaces [Schölkopf and Smola, 2002, Smola and Schölkopf, 2004]. They provide a framework for understanding the similarity measures that underpin many learning algorithms. Recent theoretical studies have increasingly focused on analyzing neural networks from the perspective of kernels. The Neural Tangent Kernel (NTK) [Jacot et al., 2018] is a seminal work that provided a way to analyze the training dynamics of infinitely wide neural networks using kernel methods. This perspective has been further explored in various contexts, leading to significant advances in our understanding of neural networks [Lee et al., 2019, Arora et al., 2019, Yang, 2019].

Despite these advances, an important question remains unexplored: *How does the similarity between hidden layer representations evolve across layers, and how is that affected by particular choices of nonlinear functions?* Previous work has mainly focused on local behaviors or specific initialization conditions [Saxe et al., 2013, Schoenholz et al., 2017, Pennington et al., 2017]. A comprehensive global analysis of neural kernel sequence fixed points and convergence properties, particularly in the presence of nonlinear activations, is still incomplete.

Code available at: https://github.com/ajoudaki/kernel-global-dynamics

This paper addresses this gap by introducing and analyzing the evolution of kernel sequences in deep neural networks. Specifically, we consider the kernel sequence $k(h_\ell(x), h_\ell(y))$, where k denotes a similarity measure, and $h_\ell(x)$ and $h_\ell(y)$ are representations of the inputs x and y at layer ell. Understanding whether and how this sequence converges to a fixed point as the depth of the network increases is crucial for uncovering the inherent implicit biases of deep networks.

Our analysis builds upon foundational work in neural network theory and leverages mean-field theory to simplify the analysis. By considering the infinite-width limit, stochastic sequences become deterministic, allowing us to focus on the underlying dynamics without the interference of random fluctuations [Poole et al., 2016, Yang et al., 2019, Mei et al., 2019].

Contributions:

- By employing algebraic properties Hermite polynomials, we derive explicit forms of the neural kernel and identify its fixed points, leading to many elegant results.
- We demonstrate that the kernel sequence converges globally to a unique fixed point for a wide class of functions that cover all practically used activations, revealing inherent biases in deep representations.
- We extend our analysis to networks with normalization layers and residual connections, highlighting their impact on the convergence behavior.

Understanding these dynamics contributes to a deeper understanding of how depth and nonlinearity interact with one another at initialization.

2 Related works

The study of deep neural networks through the lens of kernel methods and mean-field theory has garnered significant interest in recent years. The Neural Tangent Kernel (NTK) introduced by Jacot et al. [2018] provided a framework to analyze the training dynamics of infinitely wide neural networks using kernel methods. This perspective was further expanded by Lee et al. [2019] and Arora et al. [2019], who explored the connections between neural networks and Gaussian processes.

The propagation of signals in deep networks has been studied in the works of Schoenholz et al. [2017] and Pennington et al. [2017]. Previous studies have also explored the critical role of activation functions in maintaining signal propagation and stable gradient behavior in deep networks [Hayou et al., 2019]. These studies focused on understanding the conditions required for the stable propagation of information and the avoidance of signal amplification or attenuation. However, these analyses often concentrated on local behaviors or specific conditions, leaving a gap in understanding the global evolution of representations across layers.

Hermite polynomials have been used in probability theory and statistics, particularly in the context of Gaussian processes [Williams and Rasmussen, 2006]. Although Poole et al. [2016] and Daniely et al. [2016] have utilized polynomial expansions to analyze neural networks, they do not study global dynamics in neural networks, as presented in this paper. To the best of our knowledge, the only existing work that uses Hermite polynomials in the mean-field regime to study the global dynamics of the kernel is by Joudaki et al. [2023b]. However, this study only covers centered activations, which fail to capture several striking global dynamics covered in this study.

Our work extends these foundational studies by providing an explicit algebraic framework to analyze the global convergence of kernel sequences in deep networks with nonlinear activations. Using Hermite polynomial expansions, we offer a precise characterization of the kernel map and its fixed points, contributing new insights into the implicit biases of nonlinear activations.

3 Preliminaries

This section introduces the fundamental concepts, notations, and definitions used throughout this paper. If X and Y are vectors in $\mathbb{R}^n$, we use the inner product notation $\langle X, Y\rangle_{\text{avg}}$ to denote their average inner product $= \frac{1}{n}\sum_{i=1}^n X_i Y_i$. We consider a feedforward neural network with L layers and constant layer width d. The network takes an input vector $x \in \mathbb{R}^d$ and maps it to an output vector $h^L(x) \in \mathbb{R}^d$ through a series of transformations. The hidden representations in each layer ℓ are denoted by $h^\ell(x)$. The transformation in each layer is composed of a linear transformation followed by a nonlinear activation function ϕ. We consider the multilayer perceptron (MLP), the hidden representation at layer ℓ is

given by:

$$h^\ell(x) = \phi\left(\frac{1}{\sqrt{d}} W^\ell h^{\ell-1}(x)\right), \qquad W^\ell \in \mathbb{R}^{d\times d}, \tag{1}$$

where $h^0(x) = x$ is identified with the input. We can assume that elements of W^ℓ are drawn i.i.d. from a zero mean distribution, unit variance distribution. Two prominent choices for weight distributions, Gaussian $N(0,1)$ and uniform distribution Uniform$[-1,1]$, satisfy these conditions. In some variations of the MLP, we will use normalization layers and residual connections. Finally, the neural kernel between two inputs x and y at layer ℓ is defined as:

$$\rho_\ell = \langle h^\ell(x), h^\ell(y)\rangle_{\text{avg}}\,. \tag{2}$$

The main goal of this paper is to analyze the sequence $\{\rho_\ell\}$ at random initialization. The motivation for this analysis is to understand if architectural choices, specifically the activation function and the model depth, lead to specific biases of the kernel towards a certain fixed point. Namely, if there is a bias towards zero, it would imply that at initialization, the representations become more orthogonal. In contrast, a positive fixed point would mean the representations become more similar.

In the current setup, the sequence $\{\rho_\ell\}$ is a stochastic sequence or a Markov chain due to the random weights at each layer. However, we will show that under the mean-field regime, the sequence $\{\rho_\ell\}$ converges to a deterministic sequence, and we will analyze its properties.

4 The mean-field regime

In this section, we conduct a mean-field analysis of MLP to explore the neural kernel's fixed point behavior as the network depth increases. This approach allows us to gain insight into the global dynamics of neural networks, mainly how the similarity between two input samples evolves as they pass through successive network layers. Now, we can state the mean-field regime for the kernel sequence, stating that in this regime, the sequence becomes deterministic.

As will be proven later, this sequence's deterministic transition follows a scalar function defined below.

Definition 1. Given two random variables X, Y with covariance ρ, and activation ϕ, define the *kernel map* κ as the mapping between the covariance of preactivation and the covariance of post-activations:

$$\kappa(\rho) := \mathbb{E}\,\phi(X)\phi(Y), \qquad \begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right). \tag{3}$$

This definition has appeared in previous studies, namely the definition of dual kernel by Daniely et al. [2016], and has been adopted by Joudaki et al. [2023b] to study the effects of nonlinearity in neural networks.

The following proposition formally states that in the mean-field regime, the kernel sequence follows iterative applications of the kernel map.

Proposition 1. *In the mean-field regime with $d \to \infty$, let ρ_ℓ denote the kernel sequence of an MLP with activation function ϕ obeying $\mathbb{E}\,\phi(X)^2 = 1$ for $X \sim \mathcal{N}(0,1)$. If each element of the weights drawn i.i.d. from a distribution with zero mean and unit variance, the kernel sequence evolves deterministically according to the kernel map*

$$\rho_{\ell+1} = \kappa(\rho_\ell),$$

where the initial value ρ_0 corresponds to the input, and κ is defined in Definition 1.

Historically, conditions similar to $\mathbb{E}\,\phi(X)^2 = 1$ have been used to prevent quantities on the forward pass from vanishing or exploding. For example, in a ReLU, half of the activations will be zeroed out, which will lead to a vanishing norm of forward representations. The initialization of [He et al., 2016] addresses that by scaling weights to maintain consistent forward pass norms across layers. This principle is further refined by Klambauer et al. [2017], proposing the idea of self-normalizing activation functions, which ensure consistent mean and variances between pre- and post-activations.

An interesting observation is that kernel dynamics is governed by kernel map κ, defined on the basis of Gaussian preactivation, even if the weights are not Gaussian matrices. The main insight for this equivalence is that as long as elements are drawn i.i.d. from a zero mean unit variance distribution, we can apply the Central Limit Theorem (CLT) to conclude that preactivations follow the Gaussian distribution. This simple yet elegant observation gives us a powerful analytic tool to study kernel dynamics by leveraging algebraic properties for Gaussian preactivations; we will automatically get the same results for a wide class of distributions.